

Consequences of Future Data Center Deployment in Canada on Electricity Generation and Environmental Impacts

A 2015–2030 Prospective Study

Thomas Dandres, Nathan Vandromme, Glasha Obrekht, Andy Wong, Kim Khoa Nguyen, Yves Lemieux, Mohamed Cheriet, and Réjean Samson

Keywords:

consequential prospective LCA
industrial ecology
information and communications
technology (ICT)
life cycle assessment (LCA)
marginal electricity
partial equilibrium energy model



Supporting information is linked
to this article on the JIE website

Summary

The environmental impacts of data centers that provide information and communication technologies (ICTs) services are strongly related to electricity generation. With the increasing use of ICT, many data centers are expected to be built, causing more absolute impacts on the environment. Given that electricity distribution networks are very complex and dynamic systems, an environmental evaluation of future data centers is uncertain. This study proposes a new approach to investigate the consequences of future data center deployment in Canada and optimize this deployment based on the Energy 2020 technoeconomic model in combination with life cycle assessment methodology. The method determines specific electricity sources that will power the future Canadian data centers and computes related environmental impacts based on several indicators. In case-study scenarios, the largest deployment of data centers leads to the smallest impact per megawatt of data centers for all of the environmental indicators. It is found that an increase in power demand by data centers would lead to a reduction in electricity exports to the United States, driving the United States to generate more electricity to meet its energy demand. Given that electricity generation in the United States is more polluting than in Canada, the deployment of data centers in Canada is indirectly linked to an increase in overall environmental impacts. However, though an optimal solution should be found to mitigate global greenhouse gas emissions, it is not clear whether the environmental burden related to U.S. electricity generation should be attributed to the Canadian data centers.

Introduction

Information and communication technologies (ICTs) are increasingly used and becoming ubiquitous in social infrastructures (DNV 2014). While providing useful services, ICTs are also responsible for a growing share of global electricity consumption (over 4% of the world's electricity consumption in 2012) (Van Heddeghem et al. 2014) and the subsequent impacts on the environment, including climate change. ICT was

estimated to contribute around 2% of anthropogenic greenhouse gas (GHG) emissions in 2011 (Global e-Sustainability Initiative and The Boston Consulting Group 2012). Among the ICT infrastructures and equipment, data centers represent roughly one third of the total ICT electricity consumption (Van Heddeghem et al. 2014). Unlike end-user devices and transmission networks, which are spatially widely spread, data centers have a very localized power demand and are very energy intensive. The environmental impacts of electricity generation

Address correspondence to: Thomas Dandres, CIRAI, 3333 Queen-Mary, Office 310, Montréal (QC), Canada, H3C 3A7. Email: thomas.dandres@polymtl.ca

© 2016 by Yale University
DOI: 10.1111/jiec.12515

Editor managing review: Alexis Laurent

Volume 21, Number 5

depend on the type of power generation technology, and each region uses a different combination of technologies to generate electricity. Thus, the environmental impacts per kilowatt-hour (kWh) differ from one region to the next (Laurent and Espinosa 2015). Given that electricity consumption is an important source of environmental impacts in the life cycle of a data center (Arushanyan et al. 2014), its environmental footprint is highly correlated to its geographical location. Therefore, to properly evaluate the environmental footprint of data centers, it is critically important to identify the sources of electricity that power them. Whereas it may be simple (when data are available) to model the technology mix generating electricity in a specific region at the present time (Maurice 2015; Dandres et al. 2014b), it is more challenging to predict the real future electricity production. Indeed, electric networks are in constant evolution to face future power demands. New power plants are regularly added to the electric grid while old ones are decommissioned. Moreover, electric networks are interconnected, and interconnection capacities may also change over time depending on import and export needs. Also, with the threat of climate change, renewable energy sources are increasingly being integrated into electric grids. Consequently, environmental impacts per kWh of electricity consumed in a region are expected to change significantly in the future. Finally, given that data centers are very energy intensive and their global power demand keeps growing, they may have an impact on the future evolution of electric grids. This article therefore presents a methodology to evaluate environmental impacts related to long-term changes in regional power demand. This innovative methodology is based on the sequential use of a technoeconomic model of the North American energy sector with life cycle assessment (LCA) methodology. This methodology is illustrated in a case study that addresses the environmental impacts of future data centers deployed in Canada and supplied by the electric grid. In this case study, the methodology is used to investigate and optimize the future deployment of data centers in Canada by minimizing environmental impacts related to electricity generation in Canada and the United States.

Presentation of the Case Study

Canada is an attractive region for data centers: The price of electricity is lower than in many other countries (Shrinkthatfootprint 2014), around two thirds of electricity is generated from renewable energy (IEA 2015), and available cold air may be used for free cooling (The Green Grid 2012). The last two arguments have become more important in the recent year because of the increasing environmental pressure on data centers. Therefore, an acceleration of data center installation in Canada in the coming years can be anticipated. The change is already perceptible, given that Ericsson (McNevin 2013), OVH (Miller 2014), Microsoft (Sverdlik 2015), and Amazon (Barr 2016) all recently announced their plans to implement large data centers in Canada. However, this new trend is not necessarily represented in business as usual (BAU) scenarios of energy demand. Thus, this case study assesses the

environmental impacts of electricity generation in the context of an accelerating power demand by Canadian data centers in the 2015–2030 period. In Canada, the electricity consumed by data centers represents approximately 1% of the national electricity consumption (Natural Resources Canada 2016; DCD Intelligence 2013). This case study explores situations in which the Canadian data centers' demand would reach up to 2% of the national electricity demand by 2030 instead of remaining around 1%. It is anticipated that the consequences on the electric grid of deploying additional data centers are not linear. Therefore, several prospective data center deployment scenarios were studied. These five scenarios assume different future power demands of additional data centers: from 30 to 750 megawatts (MW) in 2030, as presented in figure 1.

These scenarios were developed in the context of a BAU scenario representing the global evolution of the Canadian energy demand by 2030. The BAU scenario was initially developed by the Canadian government (Environment Canada 2013) and considers an increase in the power demand of the “information and cultural industries” (that includes Canadian data centers [Statistics Canada 2007]) of 19% (from 8.0 to 9.5 terawatt-hours [TWh]) over 2015–2030 with an increase in national demand of 29% in the same period. The five scenarios assume more-significant data center power demand growths resulting in a greater increase in the information and cultural industries demand: 22%, 24%, 30%, 40%, and 75% over 2015–2030. Other aspects of the BAU scenario are presented in the *Energy 2020* section. The resulting electric grid mix of each scenario was modeled using a technoeconomic model, as presented in the *Method* section.

It is assumed that the data centers are built in the Canadian provinces with the greatest gross domestic product (GDP), population, and electric power generation: Ontario, Québec, and Alberta (Statistics Canada 2013). Considering that Ontario's GDP is roughly twice as high as those for Quebec and Alberta (which are quite similar), it was assumed that 50% of the data centers are built in Ontario, 25% in Québec, and 25% in Alberta. These provinces have very different electric grid mixes: Ontario uses mainly nuclear power, hydroelectricity, and natural gas; Québec uses almost exclusively hydroelectricity, and Alberta mostly generates electricity from coal and natural gas (Statistics Canada 2014). In terms of the overall electricity consumed per data center, it was considered that the data centers operate constantly at 66% of their capacity, like the high-end servers in Koomey (2007).

Method

This study combines LCA methodology with the use of the *Energy 2020* technoeconomic model.

Life Cycle Assessment

LCA is a method designed to study the environmental impacts of products and services based on their life cycles. The methodology is defined in International Organization for

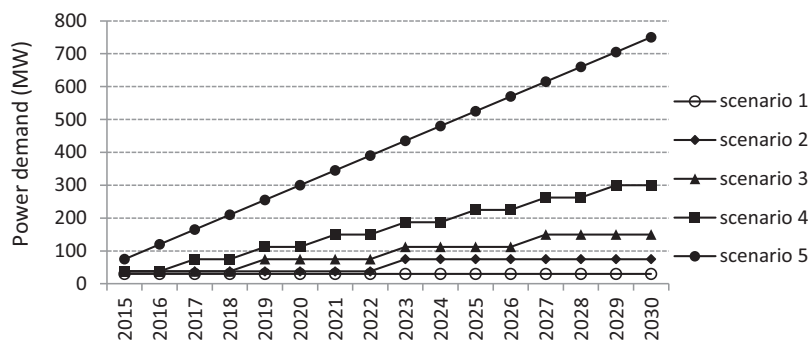


Figure 1 Scenarios of future power demands of additional data centers. MW = megawatts.

Standardization (ISO) standards (ISO 14040–14044) and remains under development given that new products and new situations create new methodological needs. It is possible to distinguish between attributional (ALCA) and consequential LCA (CLCA) approaches, which have different objectives. The ALCA objective is to evaluate the environmental impacts of a product or service over its entire life cycle at a given time (Rebitzer et al. 2004). The CLCA objective is to analyze the environmental impacts resulting from a change in the life cycle of a product or service (Ekvall and Weidema 2004; Weidema 2003; Weidema et al. 1999). In this study, the CLCA approach was followed.

Goal and Scope

The objective of this CLCA is to model the environmental impacts of the use phase of new Canadian data centers built in the future. It was therefore necessary to identify the marginal sources of electricity, which may be defined as the sources of electricity that are exclusively affected by the additional Canadian data centers. Concretely, the marginal electricity is identified by the technoeconomic model (see the *Energy 2020* section).

The studied system is the entire North American energy sector because Canada is a net exporter of electricity to the United States, and it is anticipated these exports could be affected by power demands of future Canadian data centers. The functional unit is to provide electricity each year from 2015 to 2030 to the additional data centers, as defined in each prospective scenario (figure 1).

Inventory of Elementary Flows

To compute the inventory of elementary flows corresponding to the functional unit, it is necessary to identify the electric sources that are expected to power the extra data centers in each scenario. As previously stated, electric networks are complex systems in constant evolution and their modeling in LCA is problematic (Astudillo et al. 2015; Lund et al. 2010; Messagie et al. 2014; Roux et al. 2016; Soimakallio et al. 2011; Weber et al. 2009, 2010). Determining the electricity sources affected by changes in electricity demand remains challenging in CLCA because of the uncertainty of the CLCA approach (Mathiesen

et al. 2009). Indeed, the identification of the affected technologies in CLCA is usually based on current supply-and-demand elasticities and market trends. However, the identification process is uncertain because it can hardly capture the complexity of the real markets or the sudden changes they may face in the future. Moreover, the uncertainty of the identification process is greater when the temporal horizon of the study increases. An alternative approach to CLCA is to model market effects with equilibrium economic models (Ekvall 2002). This approach has been adopted to study several topics: agriculture affected by biofuel demand (Kloverpris 2008; Kloverpris et al. 2010; Rajagopal 2014); forestry affected by bioenergy demand (Earles et al. 2013); or energy generation under different scenarios (Igos et al. 2015; Pietrapertosa et al. 2009; Eriksson et al. 2007). Although it is possible to make raw assumptions about the future of Canadian electricity generation over 2015–2030, these assumptions would lead to uncertain results and hardly allow for the identification of specific power sources in each Canadian data center deployment scenario. An economic model was therefore preferred, the details of which are presented here.

Energy 2020

The Energy 2020 energy model (Systematic Solutions 2014) was chosen because it appears especially promising for a Canadian study. Details about the model are provided in the Supporting Information available on the Journal's website.

The evolution of the global economy and demographics in 2015–2030 as well as the 2013 energy policies and climate-change mitigation plan are considered in Energy 2020 simulations through the use of a prospective scenario (the BAU scenario mentioned in the previous section) that includes assumptions on future economy. These assumptions are presented in table S1 in the supporting information on the Web. This BAU scenario already considers the evolution of the Canadian ICT sector. Therefore, the five data center deployment scenarios described in the previous section represent power demands that are independent and additional to the BAU scenario. The data centers are assumed to be entirely powered by the electric grid.

To implement the additional power demand associated with the new data centers in each scenario, the *capital energy*

requirement variable of the *data processing, hosting and related services* economic sector was adjusted exogenously in Energy 2020. The exogenous power demands of the data centers in each scenario are provided in table S2 in the supporting information on the Web. Six simulations were conducted: one for the BAU scenario and five for the case-study scenarios.

The outputs of each simulation are the quantities of energy generated each year in each region by each energy source (aggregated by technology). These outputs are reproduced in table S3 in the supporting information on the Web. The electric sources affected by data center deployment in each scenario were then evaluated by comparing the simulation results of each scenario with the simulation results of the BAU scenario. Given that economic and demographic growths are already accounted for in all scenarios, this approach makes it possible to isolate the impacts of the data centers on the electric sources. The amounts of electricity generated by each technology affected by data center deployment (i.e., the marginal electricity) in each scenario are presented in table S4 in the supporting information on the Web.

Mapping ecoinvent with Energy 2020

The affected technologies identified by Energy 2020 were modeled with the ecoinvent Database (version 3.2) (Weidema et al. 2013) and SimaPro (version 8.2.0.0; <https://simapro.com>). The list of ecoinvent processes used to model electricity generation is presented in table S5 in the supporting information on the Web. The model version of Energy 2020 used in this study has seven technologies (biomass, coal, heavy fuel oil, diesel and gasoline, hydro, natural gas, and other). When several ecoinvent processes corresponded to a single technology of Energy 2020, a weighted average process was developed (based on ecoinvent data) on the basis of the regional electricity generated by each ecoinvent process (see fraction column in table S5 in the supporting information on the Web).

Energy 2020 and ecoinvent model Canadian electric technologies at the provincial level. Thus, the ecoinvent processes were used directly for Canada. For the United States, ecoinvent provides data sets of networks regulated by the North American Electric Reliability Corporation (NERC). These networks are, however, aggregated in Energy 2020 (in the model version used in this study). Consequently, seven weighted average U.S. processes corresponding to the Energy 2020 technologies were developed from the ecoinvent data sets. The weighting was made on the basis of the amount of electricity generated (based on ecoinvent data) by each technology in each NERC region (excluding Hawaii and Alaska networks).

Technology Innovation

Modeling of technology innovation is described in the Supporting Information on the Web.

Environmental Impact Assessment

The environmental impacts were computed using SimaPro and the IMPACT2002+ method (Jolliet et al. 2003) based on

the amounts of electricity generated by each affected technology identified by Energy 2020 in the five scenarios (see table S4 in supporting information on the Web). In this article, only the endpoint indicators are presented in the *Results* section: *human health* expressed in disability-adjusted life years (DALY), *ecosystems* expressed in potentially disappeared fraction species per square meter per year (PDF/m²/yr), *resources* in megajoules (MJ) of primary energy needed to extract future resources (MJ primary) and *climate change* in kilograms of carbon dioxide equivalent (kg CO₂-eq).

Results

Figure 2 shows the annual GHG emissions related to marginal electricity consumed by additional Canadian data centers. As expected, the larger the number of new data centers, the higher the GHG emissions given that more electricity is consumed. The trend is also observed for the other impact categories: human health, ecosystems, and natural resource depletion.

A comparison of the scenarios based on the average impacts per marginal kWh of electricity consumed by data centers in 2015–2030 is illustrated in figure 3. Given that the impact categories use different units, the results are expressed as percentages in order to be included in the same figure. In an impact category, 100% is attributed to the scenario with the greatest impact for the category. Thus, the comparison of the impacts between scenarios is relative. Absolute numbers for each impact category are provided in table 1.

Figure 3 shows that the impacts on climate change and human health per marginal kWh decrease when data center deployment increases. This trend is also observed for the ecosystem and resource indicators when scenario 1 is compared to scenario 5, but:

- Scenarios 4 and 5 have more impacts per marginal kWh on the ecosystem indicator than scenario 3; and
- Scenarios 2, 3, and 4 have a little more impacts per marginal kWh on the resource indicator than scenario 1.

In other words, the impacts per MW of data center installed tend to decrease when data center deployment rises. This result is explained by the changes in the relative contribution of the marginal technologies used to generate electricity for the additional data centers (see figure 4).

In figure 4, it appears that natural gas and coal are the main marginal sources of electricity. The data centers deployed in the scenarios are therefore expected to be powered by these two sources in particular, whereas wind or biomass contribute very little to the data centers' needs. It should be noted that marginal sources of electricity differ significantly from current sources of electricity in Canada: hydro (60%), nuclear (15%), natural gas (11%), and coal (10%) (IEA 2012). Consequently, the marginal GHG emissions intensities in table 1 are high (0.89 to 1.01 kg CO₂-eq/kWh) as compared to the Canadian emissions

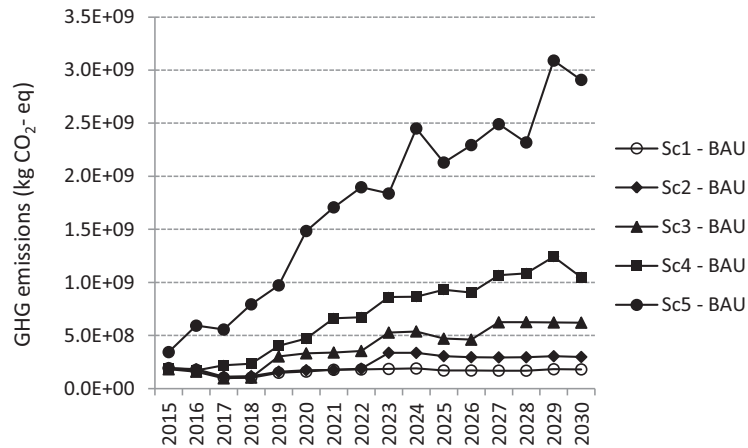


Figure 2 Greenhouse gas emissions per year and per scenario as compared to the BAU scenario. BAU = business as usual; GHG = greenhouse gas; kg CO₂-eq = kilograms of carbon dioxide equivalents; Sc = scenario.

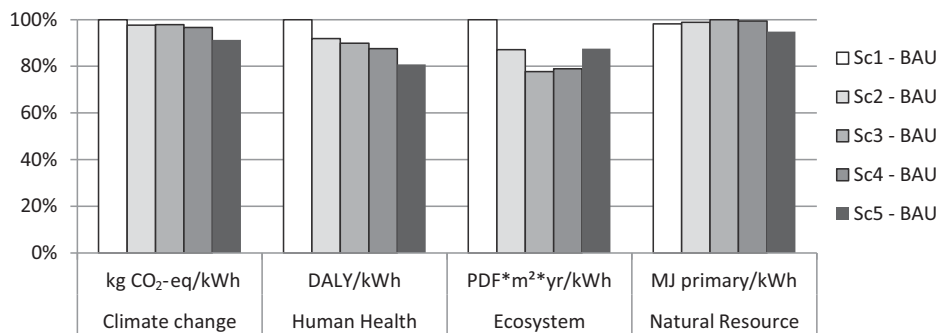


Figure 3 Relative impacts per marginal kWh, per scenario and per impact category. BAU = business as usual; DALY = disability-adjusted life years; kg CO₂-eq = kilograms of carbon dioxide equivalents; kWh = kilowatt-hour; MJ = megajoules; PDF/m²/yr = potentially disappeared fraction per square per year; Sc = scenario.

Table 1 Environmental impacts per marginal kWh, per scenario and per impact category

Scenario	Climate change kg CO ₂ -eq/kWh	Human health 10 ⁻⁶ DALY/kWh	Ecosystem PDF/m ² /yr/kWh	Natural resource MJ primary/kWh
Sc1: BAU	0.93	2.09	0.120	12.1
Sc2: BAU	0.90	1.92	0.105	12.1
Sc3: BAU	0.91	1.87	0.094	12.3
Sc4: BAU	0.90	1.83	0.095	12.2
Sc5: BAU	0.85	1.69	0.105	11.6

Note: kWh = kilowatt-hour; Sc = scenario; BAU = business as usual; kg CO₂-eq = kilograms of carbon dioxide equivalents; DALY = disability-adjusted life years; PDF/m²/yr = potentially disappeared fraction per square per year; MJ = megajoules.

intensity for electricity (0.16 kg CO₂-eq/kWh in 2013, according to Environment Canada [2015]). The marginal electricity mix is the result of different trends:

1. Hydro dams and nuclear power plants have high building costs, which are hardly justified by the small power demand of the additional data centers (considering total Canadian power demand). Moreover, these power plants take time to build. Therefore, in Energy 2020, natural gas thermal power plants are mostly built to provide extra electricity to Canadian data cen-

ters (Canadian policies discourage building coal power plants).

2. A large part of the electricity (60% to 70% depending on the scenario; percentages decrease with data center deployment) used to power the additional data centers is not produced in new power plants in Canada, but rather sourced from existing power plants by reducing electricity exports to the United States. Therefore, the United States must compensate for this reduction in its supply sources. Energy 2020 models the compensation as mainly the construction of new natural gas and coal

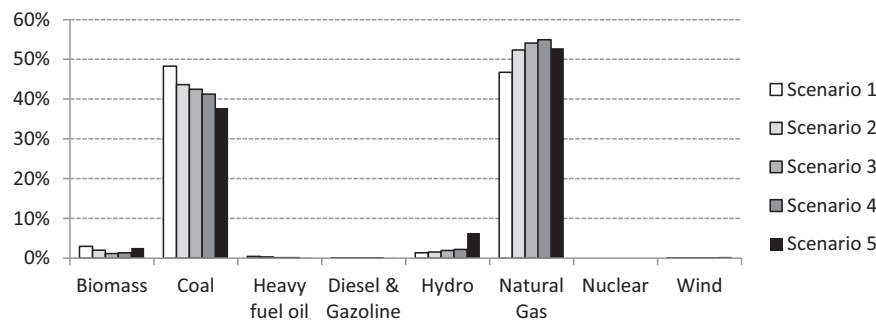


Figure 4 Relative contribution of marginal sources of electricity by scenario (2015–2030).

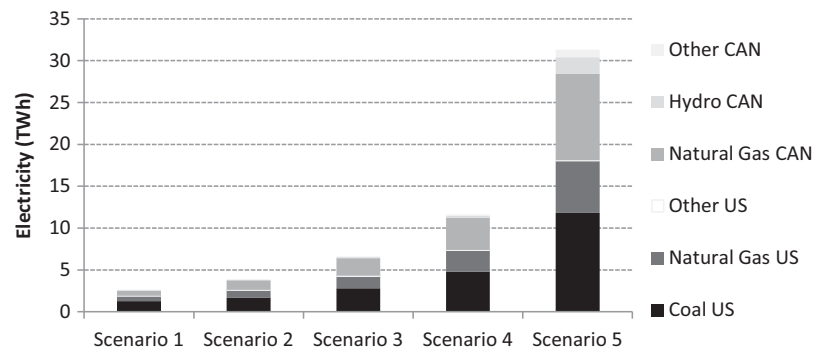


Figure 5 Marginal sources of electricity by region (2015–2030). CAN = Canada; US = United States; TWh = terawatt-hours.

thermal power plants in the United States (see marginal electricity by source aggregated over 2015–2030 in figure 5).

Results also demonstrate that contribution of natural gas to marginal electricity sources increases (from 0.47% in scenario 1 to 0.53% in scenario 5) whereas the relative contribution of coal decreases (from 0.48% in scenario 1 to 0.38% in scenario 5) when data center power demand rises. The reduction in the natural gas contribution in scenario 5 (0.53%), as compared to scenarios 3 (0.54%) and 4 (0.55%), is attributed to the increase in hydropower in Quebec (after 2020). This result typically reflects the nonlinearity of the energy system's response to an increase in the power demand of Canadian data centers.

In terms of the environmental impacts, replacing coal with natural gas in marginal electricity generation to meet an increase in data center power demand results in a decrease in human health, climate change, and ecosystems impacts per kWh given that coal power plants have a worse score than natural gas power plants for these environmental indicators. Effect on resource is less noticeable given that coal and natural gas power plants have similar scores for this indicator. The increase in bioenergy use explains the impact rise on ecosystem in figure 3 (see scenarios 4 and 5). Inecoinvent, bioenergy implies the release in the environment of ashes containing metals, which leads to significant impacts on ecosystem. Still, the greater contribution of hydro power when data centers are more broadly deployed in scenario 5 curbs the impacts per kWh for three of the four indicators.

Discussion

Evaluation of Data Center Deployment Scenarios

Scenario 5 generates greater direct environmental impacts than the other scenarios (cf. figure 2), but, at the same time, it causes less impacts on all environmental indicators when it is compared to other scenarios on the basis of 1 MW of data center installed (cf. figure 3). It should be noted, however, that “climate change” and “resources” indicators are interlinked given that fossil fuels (FFs) contribute greatly to both of these indicators. Therefore, the environmental efficiency of scenario 5 is higher (when it is compared to other scenarios). Nevertheless, the complete evaluation of each scenario should involve the services provided by the data centers: Improvement, substitution, and rebound effects among human activities should be taken into account. Although these effects are out of the scope of this study, they may be important. Indeed, the deployment of new data centers can be expected to improve the penetration of ICT in society, thus allowing avoidance of more GHG emissions in the future (Global e-Sustainability Initiative and The Boston Consulting Group 2012; IEA 2014). In this situation, the GHG emissions of each scenario would be reduced, possibly leading to negative scores if emissions reductions in other sectors surpass the data center emissions. Moreover, the heat released into the atmosphere by the data centers may be significant, but was not considered in this study because of the limited scope. Nevertheless, heat recovery for other activities, such as cultivation in greenhouses, could avoid the combustion of FFs to generate heat. Considering such effects would result in more environmental benefits.

Table 2 Environmental impact per kWh

Model for 1 kWh	kg CO ₂ -eq/kWh	10 ⁻⁷ DALY/kWh	10 ⁻² PDF/m ² /yr/kWh	MJ primary/kWh
2015 (Canada)	0.16	0.7	2.6	4.9
2015 (USA+Canada)	0.60	15.5	7.8	10.1
BAU 2015–2030 (Canada)	0.16	0.7	2.7	4.7
BAU 2015–2030 (USA+Canada)	0.63	16.0	8.1	10.3
Marginal 2015–2030 (Canada)	0.57	2.1	10.3	8.8
Marginal 2015–2030 (USA+Canada)	0.85	16.9	10.5	10.4

Note: kWh = kilowatt-hour; Sc = scenario; BAU = business as usual; kg CO₂-eq = kilograms of carbon dioxide equivalents; DALY = disability-adjusted life years; PDF/m²/yr = potentially disappeared fraction per square per year; MJ = megajoules.

Still, several researchers have pointed out that the ICT technologies might not reduce environmental impacts as compared to regular technologies. Ong and colleagues (2014) and Borggren and colleagues (2013) compared face-to-face meetings with videoconference meetings and concluded that the latter are not always better than the former. Additionally, several researchers investigated the unwanted effects of ICT usage that can equal or even offset the positive effects expected by improving human activities with ICT (Takahashi et al. 2006; Ahmadi Achachlouei and Hilty 2016; Börjesson Rivera et al. 2014). If such unwanted effects were to occur, the impacts of each scenario would be more significant. Finally, the method presented in this study addresses the use phase of future data centers, but the other life cycle phases, such as the manufacture and end of life, should also be considered in the evaluation of data center deployment.

Impact of Electricity Generation

Electricity is often an important parameter in LCA (Lund et al. 2010). It is especially the case for data centers and ICT (Arushanyan et al. 2014). In this study, it is observed that the impacts per kWh of data center usage over 2015–2030 highly depend on the modeling of the electricity mix. Indeed, while we conducted a consequential prospective LCA on the North American electric mix, other approaches would lead to different results and conclusions. These approaches, based on the Energy 2020 results, are described hereafter and their results are presented in table 2:

- In the *2015 (Canada)* approach, the impacts are computed using the 2015 Canadian electric grid mix, assuming the grid mix will remain unchanged up to 2030;
- The *2015 (USA+Canada)* approach is identical excepted that the U.S. and Canadian electric mixes are both considered;
- In the *BAU 2015–2030 (Canada)* approach, the impacts are computed using the BAU evolution of the Canadian grid mix up to 2030;
- The *BAU 2015–2030 (USA+Canada)* approach is identical except that the U.S. and Canadian electric mixes are both considered;
- In the *Marginal 2015–2030 (Canada)* approach, the impacts are computed using the marginal Canadian grid mix for scenario 5; and

- In the *Marginal 2015–2030 (USA+Canada)* approach, the impacts are computed using the marginal U.S. and Canadian grid mix for scenario 5.

Table 2 indicates that the environmental impacts per kWh largely differ depending on the approach followed to compute the electric grid. In this study, the parameter affecting the most impact per kWh is the spatial border of the system (inclusion or not of USA). Then comes the choice of the attributional or consequential approach (BAU or marginal), which has a lower, but still significant, influence on the impact per kWh. Finally, the temporal horizon has less influence in the case of the ALCA approach (not computed in the case of the CLCA approach).

Choosing the right approach to compute the electric grid mix for a study depends on the objective of the study. In our study, the objective is to evaluate the consequences of building more data centers in Canada as compared to a BAU scenario. Given that the Canadian electricity exports to the United States are found to be affected by the data center deployment, the U.S. grid mix should also be included in the computing of the electric grid mix. Therefore, the Marginal 2015–2030 (USA+Canada) approach of table 2 should be used.

If the objective was to evaluate the carbon footprint of additional data centers, then the 2015–2030 emissions computed in a relevant data center deployment scenario for the United States and Canada should be used. The use of the BAU 2015–2030 (USA+Canada) approach would also be suitable if the data center power demand would remain low in comparison to the total power generation of the United States and Canada. In this case, the use of the marginal factors would not be fair given that certain low-emissions technologies, such as nuclear, would be excluded from the marginal electricity. Anyway, current legislation and guidelines for carbon footprint assessment do not consider marginal electricity, and the debate on marginal electricity allocation is unresolved.

The 2015–2030 (BAU, Canada) approach should not be used because it would not capture the U.S. side effect. Such factors would largely underestimate the impacts of the Canadian data centers. However, in the context of a tax on environmental externalities, the attribution of the U.S. environmental burden to the Canadian users is debatable and could justify the use of the 2015–2030 (BAU, Canada) approach. Nevertheless, the question of the attribution of the indirect effects of

decision making has not yet been answered in the context of environmental footprints.

Impacts factors for 2015 (Canada or USA+Canada) should not be used to study future emissions of data centers because they do not reflect the future evolution of electricity generation. To summarize, the choice of the approach should be in agreement with the geographical and temporal scopes of the study and cover anticipated indirect effects. Marginal approaches are especially relevant in the context of optimization because they reflect the nonlinearity of impacts. However, the use of marginal approaches results in attribution of different impacts between marginal and nonmarginal users. Nevertheless, such an equity problem can be overcome by combining the attributional and marginal approaches to share the marginal impacts among all users.

Contributions to Energy 2020 and Life Cycle Assessment

Combining Energy 2020 and LCA enhances both tools. On one hand, the approach adds environmental indicators based on the power plants' life cycles to the Energy 2020 results, providing more-valuable information on the potential environmental impacts of future energy policies given that more environmental indicators are considered. On the other hand, Energy 2020 makes it possible to conduct an LCA using prospective (Pesonen et al. 2000) and consequential (Ekvall and Weidema 2004) methodologies in the context of the North American energy sector. Instead of using expert judgments, extrapolating time series, making uncertain assumptions, and using simplified economic models as in standard prospective and consequential LCA, an acknowledged advanced model is used. Therefore, it is expected that the proposed method will provide more-robust results for LCA involving the future evolution of the North American energy sector. But the use of economic models in LCA is quite new (Kloverpris 2008; Earles 2011; Dandres et al. 2011), and further studies are required to illustrate the advantages and flaws of the combination.

Limitations, Uncertainty, and Recommendations

Energy 2020 is expected to be one of the best energy models to investigate changes in the North American energy sector. However, it relies on assumptions about the future evolution of macroeconomic variables and economic modeling, which cannot predict noneconomic events such as war, natural disasters, economic crisis, or a political decision. For instance, this project was achieved in 2014 at a time when the U.S. clean power plan was not announced. Thus, our simulations do not take into account the future reduction of carbon intensity in U.S. electricity generation as targeted by the U.S. clean power plan, and our results probably overestimate the marginal emissions from U.S. power plants. Additionally, there is some uncertainty on the electric mixes attributed to imperfect convergence of solutions computed by the Energy 2020 (see tables S2 and S4 in the supporting information on the Web).

Therefore, the results of the model should not be taken as is, but rather used to identify key trends and unwanted effects. In this case study, the main result may be the unexpected reduction in Canadian electricity exports to the United States and the related negative environmental effect. By being aware of the unwanted effect before it occurs, it is possible to anticipate and take appropriate measures (e.g., subsidize the generation of renewable energy to power data centers in Canada) to counter it. An iterative approach would support the design of an optimal strategy.

Regarding LCA, the limitations pertain to the quality of data used to model electricity generation and evaluate the environmental impacts. Indeed, generic processes were used to model power plants regardless of the power plants' specificities. Thus, there is uncertainty with regard to the computed emissions. Moreover, technological changes are imperfectly modeled in this LCA. Whereas future technologies are expected to be more efficient than current ones, data collected on power plants' efficiency suggest that thermal technologies are not improved. Therefore, the results may overestimate the future impacts on the environment given that natural gas and coal technologies are found to play a major role in the future. Prospective data cannot be obtained by measurement and must be forecasted. However, forecasting requires significant resources (e.g., a panel of experts [Godet et al. 2004]) that were not available at the time of the study. Also, environmental impacts were computed regardless of the geographical location of the emissions, even though the site of an emission may affect its fate in the environment and result in different environmental impacts depending on the location (Manne et al. 2010). Although, ideally, the location of the emissions should be taken into account, the regional impact assessment models in LCA are still under development. Also, it was assumed that the cause-and-effects chains that link the substances emissions to the environmental impacts would remain stable in the future. However, this assumption may overestimate the capacity of ecosystems to resist to damages and fail to represent the cases where this capacity is exceeded. Such situation could occur, for instance, in the case of acidification (Roy et al. 2012). Finally, because of technical issues and lack of resources, it was not possible to investigate a large number of scenarios, conduct sensitivity analyses, or run Monte Carlo simulations to evaluate the uncertainty of the final results. Indeed, such analyses are time- and resource-consuming when an equilibrium model such as Energy 2020 is used sequentially with LCA (Dandres et al. 2014a). Moreover, the evaluation of impact assessment uncertainty is not currently possible in SimaPro because of lack of data.

Conclusion

The environmental assessments of data centers and ICT are closely linked to electricity generation modeling. The accuracy of electricity impact data is a crucial parameter in environmental ICT studies. Therefore, it is relevant to study the future electric grid mixes that will supply future ICT systems in

order to evaluate their environmental performances. Given that electric grid mixes evolve in time, adequate tools must be used to model them properly rather than applying *ceteris paribus* or BAU assumptions. In this study, a regional economic model (Energy 2020) was used to represent the changes in energy generation in Canada and the United States in 2015–2030. Combined with the LCA methodology, the model made it possible to evaluate the environmental impacts of several data center deployment scenarios in Canada and determine those that generate the least impacts per MW of data center built. In this case study, the largest deployment of data centers (+750 MW) is the most efficient scenario for three of the four environmental indicators. A smaller deployment of data centers (+150 MW) would minimize the impact per MW of data center built on ecosystem, but would not minimize impacts on the human health and climate-change, indicators. Moreover, it would maximize the impact on the resource indicator.

The combination of a technoeconomic model and LCA methodology capitalizes on the use of both tools: The model provides prospective scenarios, and LCA computes several environmental impacts based on the technologies' life cycles. Whereas the proposed methodology has been applied to elucidate the consequences of data centers' deployment, it could also be used to evaluate changes in other industrial activities. A significant result is that the increase in the power demand of the Canadian data centers causes a reduction in electricity exports to the United States. Consequently, U.S. electricity generation must increase to compensate for the lack of electricity in the United States. Given that this compensation is achieved with coal and natural gas, major environmental impacts are indirectly generated by the deployment of Canadian data centers. However, the Clean Power Plan announced in August 2015 (after the study achievement) is expected to reduce the contribution from FFs in U.S. electricity generation. Therefore, marginal impacts from the United States might be smaller than those anticipated in this study. Moreover, whether these indirect impacts should be attributed to the Canadian data centers is debatable. Indeed, although there is a causal link between the increase in Canadian data center deployment and the decrease in electricity exports to the United States, Canadian electricity providers have no control over the choice of the fuel for the U.S. electricity generation. This issue is of major concern in terms of GHG emissions given that the climate-change issue is a global one. Therefore, guidelines must be established in order to contend with indirect effects in carbon footprint methodologies.

Acknowledgments

The authors thank the NSERC of Canada for their financial support under grant CRDPJ 424371-11, the reviewers for their pertinent comments, and Lisa Cecchini for language checking. The views expressed in this article are those of the authors and do not necessarily reflect those of Environment Canada or the Government of Canada.

References

- Ahmadi Achachlouei, M. and L. M. Hilty. 2016. Using systems thinking and system dynamics modeling to understand rebound effects. In *Advances and new trends in environmental and energy informatics: Selected and extended contributions from the 28th International Conference on Informatics for Environmental Protection*, edited by J. Marx Gomez et al. Cham, Switzerland: Springer International.
- Arushanyan, Y., E. Ekener-Petersen, and G. Finnveden. 2014. Lessons learned—Review of LCAs for ICT products and services. *Computers in Industry* 65(2): 211–234.
- Astudillo, M. F., K. Treyer, C. Bauer, and M. B. Amor. 2015. Exploring challenges and opportunities of life cycle management in the electricity sector. In *LCA compendium—The complete world of life cycle assessment*, edited by G. Sonnemann and M. Margni. Dordrecht, the Netherlands: Springer.
- Barr, J. 2016. Amazon Web Services official blog. <https://aws.amazon.com/blogs/aws/in-the-works-aws-region-in-canada/>. Accessed 27, 2016.
- Borggren, C., Å. Moberg, M. Räsänen, and G. Finnveden. 2013. Business meetings at a distance—Decreasing greenhouse gas emissions and cumulative energy demand? *Journal of Cleaner Production* 41: 126–139.
- Börjesson Rivera, M., C. Håkansson, Å. Svenfelt, and G. Finnveden. 2014. Including second order effects in environmental assessments of ICT. *Environmental Modelling & Software* 56: 105–115.
- Dandres, T., C. Gaudreault, P. Tirado-Seco, and R. Samson. 2011. Assessing non-marginal variations with consequential LCA: Application to European energy sector. *Renewable and Sustainable Energy Reviews* 15(6): 3121–3132.
- Dandres, T., C. Gaudreault, P. T. Seco, and R. Samson. 2014a. Uncertainty management in a macro life cycle assessment of a 2005–2025 European bioenergy policy. *Renewable and Sustainable Energy Reviews* 36: 52–61.
- Dandres, T., R. Farrahi Moghaddam, K. K. Nguyen, Y. Lemieux, M. Cheriet, and R. Samson. 2014b. Minimization of Telco cloud emissions based on marginal electricity management. In *ICT4S 2014*, 24–27 August, Stockholm, Sweden.
- DCD (Data Centre Dynamics Ltd.) Intelligence. 2013. *2013 census report: Global data center power 2013*. London: DCD Intelligence, Data Centre Dynamics Ltd.
- DNV. 2014. *Technology report outlook 2020*. Høvik, Norway: DNV GL AS NO-1322.
- Earles, J. M. 2011. LCA and forest biorefining: Environmental assessment of a modified OSB Mill and an integrated partial equilibrium framework for policy analysis. Thesis, Forest Resources, The University of Maine, Orono, ME, USA.
- Earles, J. M., A. Halog, P. Ince, and K. Skog. 2013. Integrated economic equilibrium and life cycle assessment modeling for policy-based consequential LCA. *Journal of Industrial Ecology* 17(3): 375–384.
- Ekvall, T. 2002. Cleaner production tools: LCA and beyond. *Journal of Cleaner Production* 10(5): 403–406.
- Ekvall, T. and B. P. Weidema. 2004. System boundaries and input data in consequential life cycle inventory analysis. *The International Journal of Life Cycle Assessment* 9(3): 161–171.
- Environment Canada. 2013. *Canada's emissions trends*. En81-18/2013E-PDF. Gatineau (QC), Canada: Environment Canada.
- Environment Canada. 2015. *National inventory report 1990–2013: Greenhouse gas sources and sinks in Canada (part 3)*. Gatineau (QC), Canada: Environment Canada.

- Eriksson, O., G. Finnveden, T. Ekvall, and A. E. Bjorklund. 2007. Life cycle assessment of fuels for district heating: A comparison of waste incineration, biomass- and natural gas combustion. *Energy Policy* 35(2): 1346–1362.
- Global e-Sustainability Initiative and The Boston Consulting Group. 2012. *GeSI SMARTer 2020*. Global e-Sustainability Initiative. Brussels, Belgium: Global e-Sustainability Initiative (GeSI).
- Godet, M., R. Monti, F. Meunier, and F. Roubelat. 2004. *La boîte à outils de prospective stratégique [Toolbox for strategic prospects]*. Laboratoire d'investigation en prospective, stratégie et organisation. Paris, France: LIPSOR.
- IEA (International Energy Agency). 2012. Electricity/heat in Canada in 2012. www.iea.org/statistics/statisticssearch/report/?country=CANADA=&product=electricityandheat&year=Select. Accessed 11 December 2012.
- IEA (International Energy Agency). 2014. *Tracking clean energy progress 2014*. Paris: IEA.
- IEA (International Energy Agency). 2015. *Energy policies of IEA countries—Canada*. Paris: IEA.
- Igos, E., B. Rugani, S. Rege, E. Benetto, L. Drouet, and D. S. Zachary. 2015. Combination of equilibrium models and hybrid life cycle-input-output analysis to predict the environmental impacts of energy policy scenarios. *Applied Energy* 145: 234–245.
- Jolliet, O., M. Margni, R. Charles, S. Humbert, J. Payet, G. Rebitzer, and R. K. Rosenbaum. 2003. IMPACT 2002+: A new life cycle impact assessment methodology. *The International Journal of Life Cycle Assessment* 8(6): 324–330.
- Klopperpris, J. H. 2008. Life cycle inventory modelling of land use induced by crop consumption—Part 1: Conceptual analysis and methodological proposal. *The International Journal of Life Cycle Assessment* 13(1): 13–21.
- Klopperpris, J. H., K. Baltzer, and P. Nielsen. 2010. Life cycle inventory modelling of land use induced by crop consumption—Part 2: Example of wheat consumption in Brazil, China, Denmark and the USA. *The International Journal of Life Cycle Assessment* 15(1): 90–103.
- Koomey, J. G. 2007. *Estimating total power consumption by servers in the US and the world*. Palo Alto, CA: Lawrence Berkeley National Laboratory, Stanford University.
- Laurent, A. and N. Espinosa. 2015. Environmental impacts of electricity generation at global, regional and national scales in 1980–2011: What can we learn for future energy planning? *Energy & Environmental Science* 8(3): 689–701.
- Lund, H., B. V. Mathiesen, P. Christensen, and J. H. Schmidt. 2010. Energy system analysis of marginal electricity supply in consequential LCA. *The International Journal of Life Cycle Assessment* 15(3): 260–271.
- Manneh, R., M. Margni, and L. Deschenes. 2010. Spatial variability of intake fractions for canadian emission scenarios: A comparison between three resolution scales. *Environmental Science & Technology* 44(11): 4217–4224.
- Mathiesen, B. V., M. Münster, and T. Fruergaard. 2009. Uncertainties related to the identification of the marginal energy technology in consequential life cycle assessments. *Journal of Cleaner Production* 17(15): 1331–1338.
- Maurice, E. 2015. *Modélisation temporelle de la consommation électrique en analyse du cycle de vie, appliquée au contexte des TIC [Temporal modelling of electricity consumption in life cycle assessment: The ICT context]*. Thesis, Département de Génie Chimique, Université de Montréal, Montréal, Québec, Canada.
- McNevin, A. 2013. Ericsson to build three giant data centers. www.datacenterdynamics.com/news/ericsson-to-build-three-giant-data-centers/81908.article. Accessed 1 April 2014.
- Message, M., J. Mertens, L. Oliveira, S. Rangaraju, J. Sanfelix, T. Coosemans, J. van Mierlo, and C. Macharis. 2014. The hourly life cycle carbon footprint of electricity generation in Belgium, bringing a temporal resolution in life cycle assessment. *Applied Energy* 134: 469–476.
- Miller, R. 2014. OVH fills massive Quebec data center with cube-shaped hosting towers. www.datacenterknowledge.com/archives/2014/01/20/ovh-data-center/. Accessed 6 January 2015.
- Natural Resources Canada. 2016. Data centres. www.nrcan.gc.ca/energy/products/categories/data-centres/13741. Accessed 26 February 2016.
- Ong, D., T. Moors, and V. Sivaraman. 2014. Comparison of the energy, carbon and time costs of videoconferencing and in-person meetings. *Computer Communications* 50: 86–94.
- Pesonen, H.-L., T. Ekvall, G. Fleischer, G. Huppes, C. Jahn, Z. S. Klos, G. Rebitzer, et al. 2000. Framework for scenario development in LCA. *The International Journal of Life Cycle Assessment* 5(1): 21–30.
- Pietrapertosa, F., C. Cosmi, M. Macchiato, M. Salvia, and V. Cuomo. 2009. Life cycle assessment, ExternE and comprehensive analysis for an integrated evaluation of the environmental impact of anthropogenic activities. *Renewable and Sustainable Energy Reviews* 13(5): 1039–1048.
- Rajagopal, D. 2014. Consequential life cycle assessment of policy vulnerability to price effects. *Journal of Industrial Ecology* 18(2): 164–175.
- Rebitzer, G., T. Ekvall, R. Frischknecht, D. Hunkeler, G. A. Norris, T. Rydberg, W. P. Schmidt, S. Suh, B. P. Weidema, and D. W. Pennington. 2004. Life cycle assessment Part 1: Framework, goal and scope definition, inventory analysis, and applications. *Environmental International* 30(5): 701–720.
- Roux, C., P. Schalbart, and B. Peuportier. 2016. Accounting for temporal variation of electricity production and consumption in the LCA of an energy-efficient house. *Journal of Cleaner Production* 113: 532–540.
- Roy, P.-O., L. Deschênes, and M. Margni. 2012. Life cycle impact assessment of terrestrial acidification: Modeling spatially explicit soil sensitivity at the global scale. *Environmental Science & Technology* 46(15): 8270–8278.
- Shrinkthatfootprint. 2014. Average electricity prices around the world: \$/kWh. <http://shrinkthatfootprint.com/average-electricity-prices-kwh>. Accessed 8 January 2015.
- Soimakallio, S., J. Kiviluoma, and L. Saikku. 2011. The complexity and challenges of determining GHG (greenhouse gas) emissions from grid electricity consumption and conservation in LCA (life cycle assessment): A methodological review. *Energy* 36(12): 6705–6713.
- Statistics Canada. 2007. North American Industry Classification System (NAICS) 2007. <http://stds.statcan.gc.ca/naics-scian/2007/ts-rt-eng.asp?criteria=51>. Accessed 23 February 2016.
- Statistics Canada. 2013. CANSIM tables. Ottawa (ON), Canada: Statistics Canada.
- Statistics Canada. 2014. Electric power generation, by class of electricity producer, edited by CANSIM. Ottawa (ON), Canada: Statistics Canada.
- Sverdlik, Y. 2015. First two Microsoft data centers coming to Canada in 2016. www.datacenterknowledge.com/archives/2015/06/03/

- first-two-microsoft-data-centers-coming-to-canada-in-2016/. Accessed 10 December 2015.
- Systematic Solutions. 2014. ENERGY 2020 overview. www.energy2020.com/resources.html. Accessed 8 January 2015.
- Takahashi, K. I., M. Tsuda, J. Nakamura, and S. Nishi. 2006. Estimation of videoconference performance: Approach for fairer comparative environmental evaluation of ICT services. Paper presented at Proceedings of the 2006 IEEE International Symposium on Electronics and the Environment, 8–11 May, San Francisco, CA, USA.
- The Green Grid. 2012. *Data center efficiency and IT equipment reliability at wider operating temperature and humidity ranges*. Beaverton, OR: The Green Grid.
- Van Heddeghem, W., S. Lambert, B. Lannoo, D. Colle, M. Pickavet, and P. Demeester. 2014. Trends in worldwide ICT electricity consumption from 2007 to 2012. *Computer Communications* 50: 64–76.
- Weber, C. L., P. Jaramillo, J. Marriott, and C. Samaras. 2009. Uncertainty and variability in accounting for grid electricity in life cycle assessment. Paper presented at the 2009 IEEE International Symposium on Sustainable Systems and Technology (ISSST'09), 18–20 May, Tempe, AZ, USA.
- Weber, C. L., P. Jaramillo, J. Marriott, and C. Samaras. 2010. Life cycle assessment and grid electricity: What do we know and what can we know? *Environmental Science & Technology* 44(6): 1895–1901.
- Weidema, B. P. 2003. Market information in life cycle assessment, edited by Danish Environmental Protection Agency. Copenhagen, Denmark: Ministry of environment and food of Denmark.
- Weidema, B. P., N. Frees, and A.-M. Nielsen. 1999. Marginal production technologies for life cycle inventories. *The International Journal of Life Cycle Assessment* 4(1): 48–56.
- Weidema, B. P., C. Bauer, R. Hischier, C. Mutel, T. Nemecek, J. Reinhard, C. O. Vadenbo, and G. Wernet. 2013. *Overview and methodology. Data quality guideline for the ecoinvent Database version 3. Ecoinvent report 1 (v3)*. St. Gallen, Switzerland: The ecoinvent Centre.

About the Authors

Thomas Dandres is a research officer at CIRAIG (Polytechnique Montréal) and Synchromedia (École de Technologie Supérieure) in Montréal (QC, Canada). **Nathan Vandromme** is a master student (graduated) at CIRAIG (Polytechnique Montréal) and Synchromedia (École de Technologie Supérieure). **Glasha Obrekht** is a senior economist at Environment and Climate Change Canada in Gatineau (QC, Canada). **Andy Wong** was a senior analyst at Environment and Climate Change Canada in Gatineau (QC, Canada). **Kim Khoa Nguyen** is a professor at École de Technologie Supérieure, Synchromedia. **Yves Lemieux** is a senior sustainability and Storage Researcher at Ericsson Canada Inc. in Montréal (QC, Canada). **Mohamed Cheriet** is the director of Synchromedia and a professor at École de Technologie Supérieure. **Réjean Samson** is the director of CIRAIG and a Professor at Polytechnique Montreal in Montréal (QC, Canada).

Supporting Information

Supporting information is linked to this article on the JIE website:

Supporting Information S1: This supporting information includes additional information about the Energy 2020 model, the scenarios (macroeconomic changes and future electricity demands), the modeling of the regional grid mixes with ecoinvent, and the results of the Energy 2020 simulations.